AI ASSISTED CODING

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Batch: 37 **Assignment-** 14.3

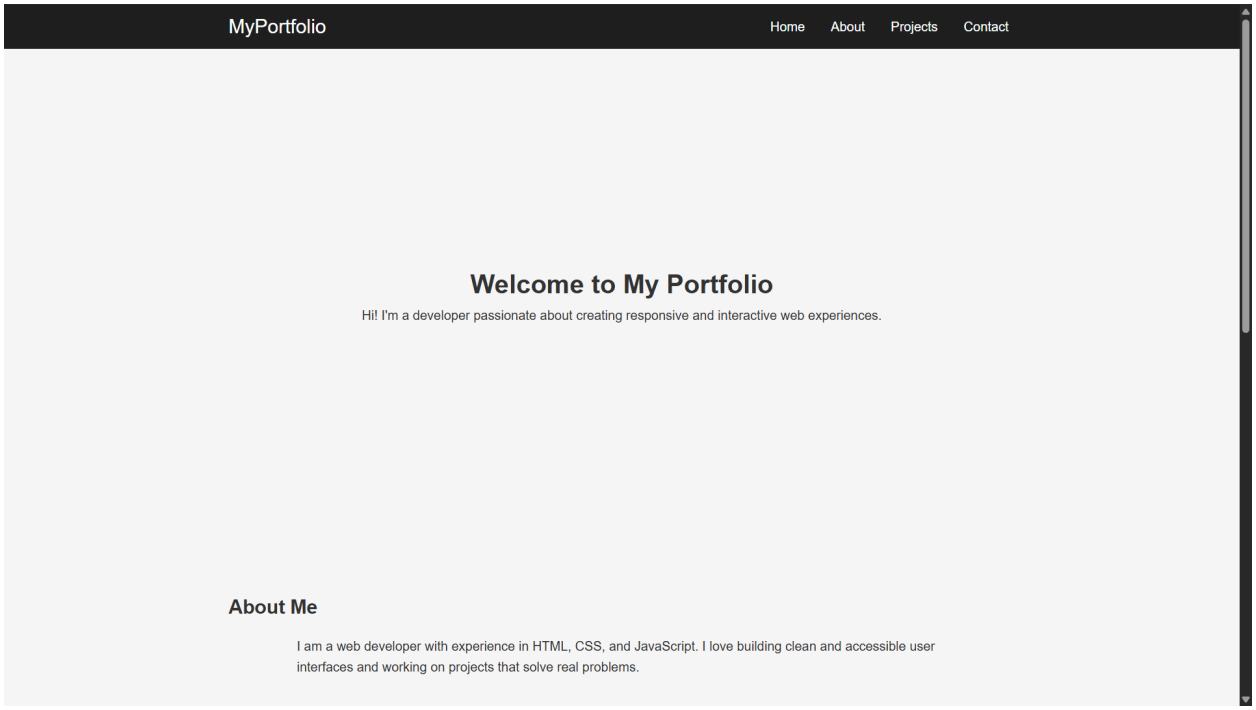
Task 1: AI-Assisted Personal Portfolio Website

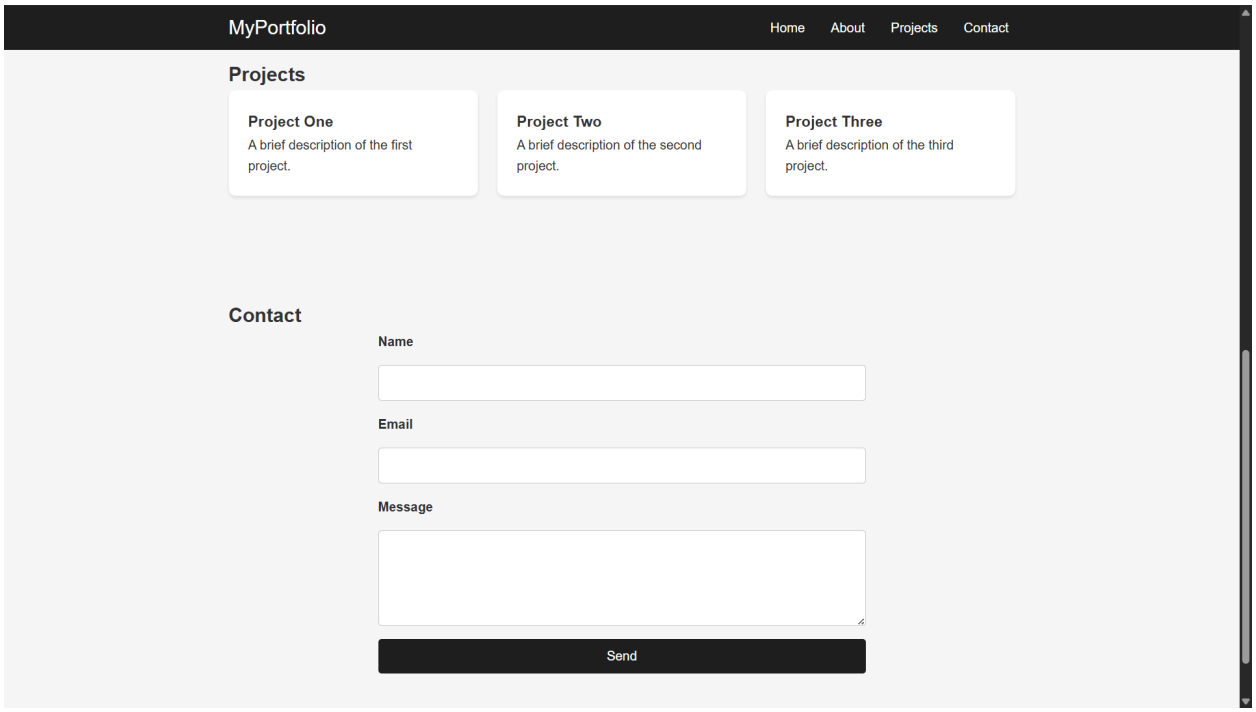
Prompt:"**#Generate a semantic and well-structured personal portfolio website using HTML5 and CSS3.**
The website should contain separate sections for Home, About, Projects, and Contact.
Use modern responsive design techniques such as Flexbox or CSS Grid so the layout adapts to desktop and mobile screens.
Add interactive UI features such as hover effects on project cards and smooth scrolling navigation links.
Separate the styling into an external CSS file to maintain clean code organization and readability.Ensure the layout is visually appealing and easy to navigate."

Index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Personal Portfolio</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <header class="site-header">
        <nav class="navbar">
            <a href="#home" class="nav-logo">MyPortfolio</a>
            <ul class="nav-links">
                <li><a href="#home">Home</a></li>
                <li><a href="#about">About</a></li>
                <li><a href="#projects">Projects</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
            <button class="nav-toggle" aria-label="toggle navigation">
                <span class="hamburger"></span>
            </button>
        </nav>
    </header>

    <main>
        <section id="home" class="section home-section">
            <div class="content">
                <h1>Welcome to My Portfolio</h1>
                <p>Hi! I'm a developer passionate about creating responsive and
                    interactive web experiences.</p>
            </div>
        </section>

        <section id="about" class="section about-section">
            <h2>About Me</h2>
            <p>
                I am a web developer with experience in HTML, CSS, and JavaScript. I love
                building clean and accessible user interfaces and working on projects that
                solve real problems.
            </p>
        </section>

        <section id="projects" class="section projects-section">
            <h2>Projects</h2>
            <div class="project-grid">
```

```html
<html lang="en">
<body>
    <main>

        <section id="contact" class="section contact-section">
            <h2>Contact</h2>
            <form class="contact-form">
                <label for="name">Name</label>
                <input type="text" id="name" name="name" required>

                <label for="email">Email</label>
                <input type="email" id="email" name="email" required>

                <label for="message">Message</label>
                <textarea id="message" name="message" rows="5" required></textarea>

                <button type="submit">Send</button>
            </form>
        </section>
    </main>

    <footer class="site-footer">
        <p>&copy; 2026 MyPortfolio. All rights reserved.</p>
    </footer>

    <script>
        // Smooth scroll behavior for navigation links
        document.querySelectorAll('a[href^="#"]').forEach(anchor => {
            anchor.addEventListener('click', function (e) {
                e.preventDefault();
                document.querySelector(this.getAttribute('href')).scrollIntoView({
                    behavior: 'smooth'
                });
            });
        });

        // Toggle mobile nav
        const toggle = document.querySelector('.nav-toggle');
        const navLinks = document.querySelector('.nav-links');
        toggle.addEventListener('click', () => {
            navLinks.classList.toggle('active');
        });
    </script>
</body>
</html>
```





Explanation:

GitHub Copilot generated a semantic HTML structure with clearly defined sections representing the portfolio content. An external CSS file was used to implement responsive styling using Flexbox for flexible layout alignment. Hover effects were applied to project cards to improve interactivity, and

smooth scrolling was implemented for navigation links to create a better user experience.

Output:

The final portfolio webpage contains:
index.html → HTML structure of the portfolio website
style.css → External CSS styling for layout and responsiveness
The page includes a navigation bar linking to Home, About, Projects, and Contact sections.Project cards display with hover animations, and the layout adjusts correctly for different screen sizes.

Observation:

AI-assisted code generation helped quickly create a complete portfolio layout while maintaining clean code structure.Minor manual adjustments were required to improve spacing and responsiveness.

Task 2: AI-Generated Restaurant Landing Page

Prompt: "#Generate a responsive restaurant landing page using HTML, CSS, and JavaScript.
The page should include a navigation bar with links to Home, Menu, Gallery, and Contact sections.
Create a visually appealing menu section using CSS card layouts to display food items and descriptions.
Implement a JavaScript image slider for the gallery section with smooth transitions between images.
Use an external CSS file for styling and add simple animations or hover effects to enhance user experience.
Ensure the layout remains responsive across different screen sizes."

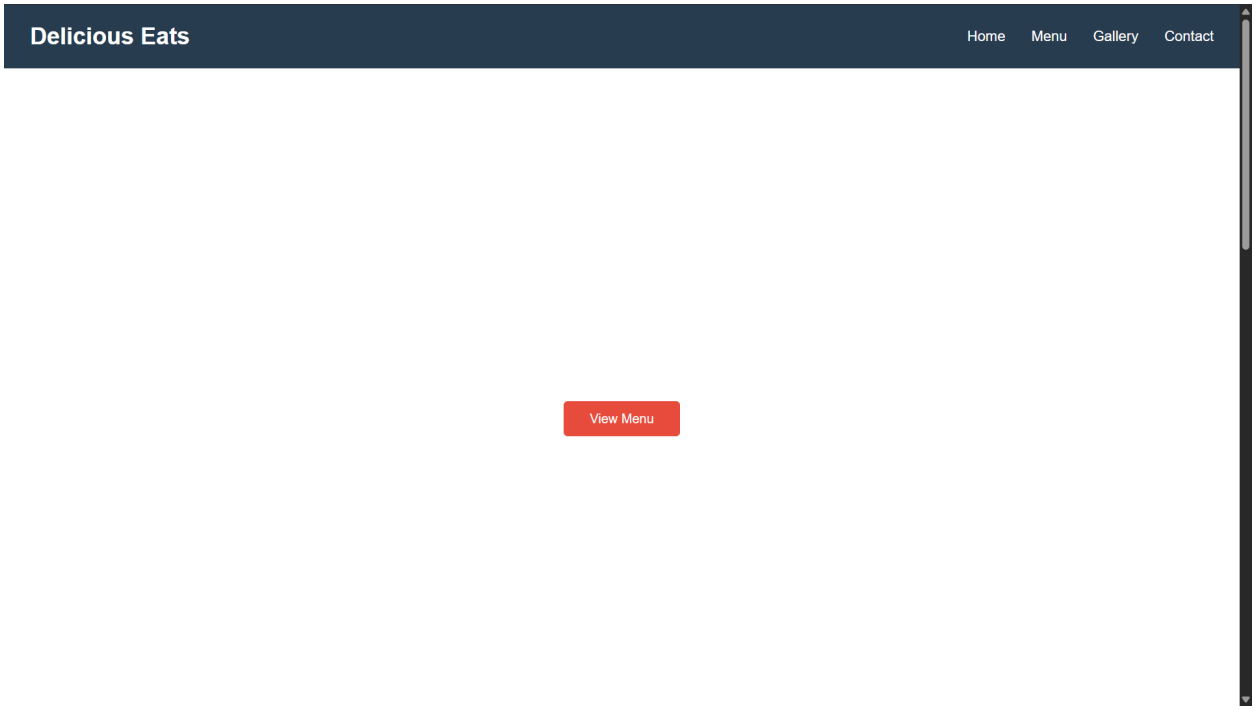
restaurant.html

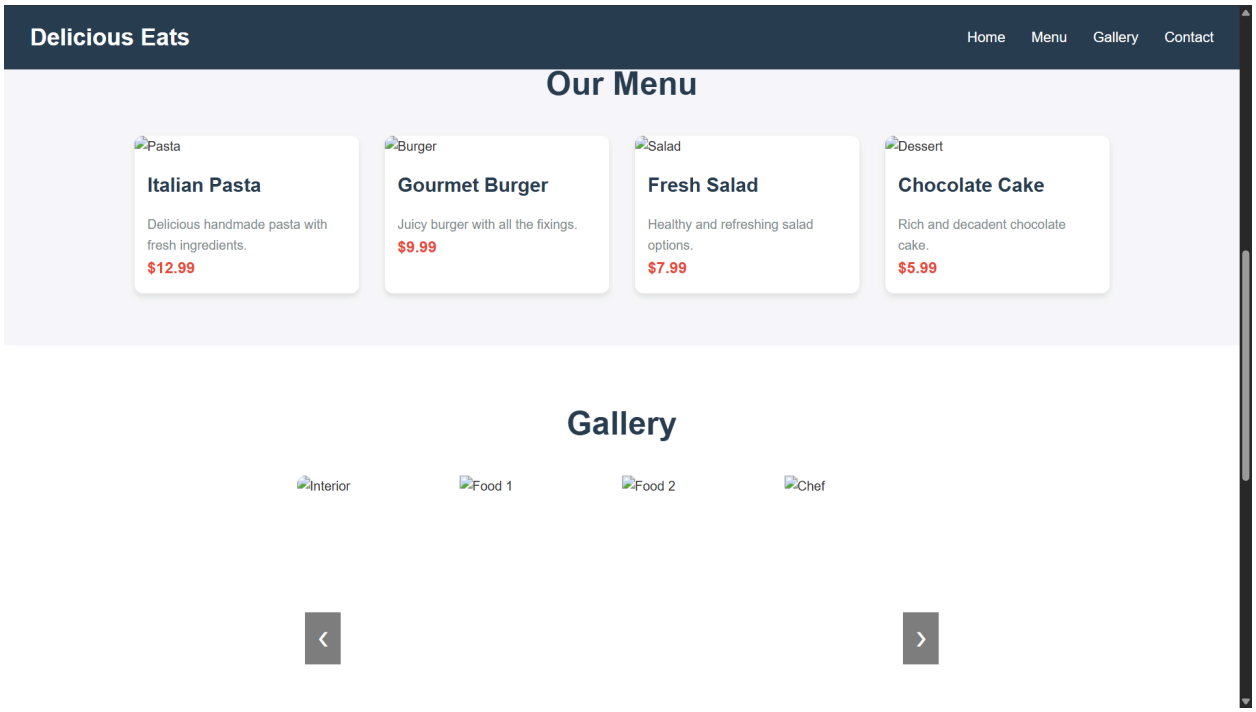
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Landing Page</title>
    <link rel="stylesheet" href="restaurant.css">
</head>
<body>
    <header>
        <nav class="navbar">
            <div class="logo">Delicious Eats</div>
            <ul class="nav-links">
                <li><a href="#home">Home</a></li>
                <li><a href="#menu">Menu</a></li>
                <li><a href="#gallery">Gallery</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
            <div class="hamburger" id="hamburger">
                <span></span>
                <span></span>
                <span></span>
            </div>
        </nav>
    </header>

    <section id="home" class="hero">
        <div class="hero-content">
            <h1>Welcome to Delicious Eats</h1>
            <p>Experience the finest cuisine in town.</p>
            <a href="#menu" class="btn">View Menu</a>
        </div>
    </section>

    <section id="menu" class="menu">
        <h2>Our Menu</h2>
        <div class="menu-grid">
            <div class="menu-card">
                <img src="https://via.placeholder.com/300x200?text=Pasta" alt="Pasta">
                <h3>Italian Pasta</h3>
                <p>Delicious handmade pasta with fresh ingredients.</p>
                <span class="price">$12.99</span>
            </div>
            <div class="menu-card">
                <img src="https://via.placeholder.com/300x200?text=Burger" alt="Burger">
```

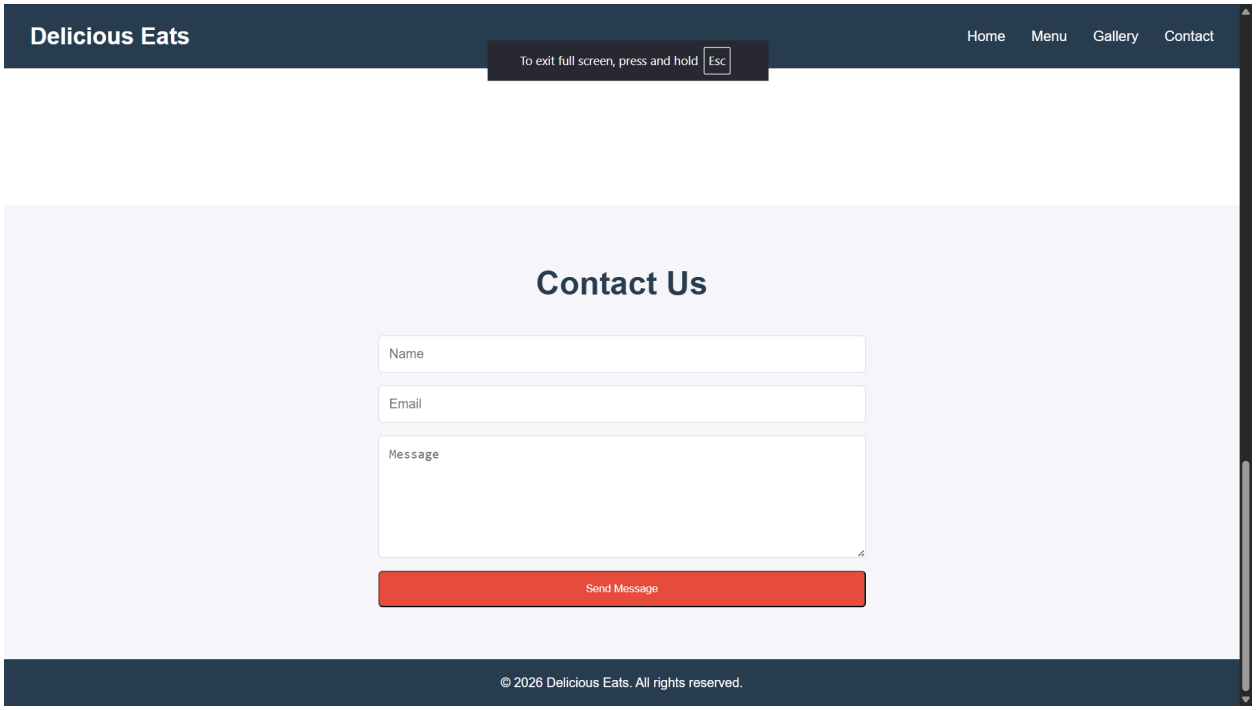
```css
/* Reset and base styles */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Arial', sans-serif;
    line-height: 1.6;
    color: #333;
    background-color: #fff;
}

/* Navigation */
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background-color: #2c3e50;
    padding: 1rem 2rem;
    position: fixed;
    width: 100%;
    top: 0;
    z-index: 1000;
}

.logo {
    color: #fff;
    font-size: 1.8rem;
    font-weight: bold;
}

.nav-links {
    list-style: none;
    display: flex;
}

.nav-links li {
    margin-left: 2rem;
}

.nav-links a {
    color: #fff;
    text-decoration: none;
```

```
268     .nav-links {
269         display: none;
270         flex-direction: column;
271         position: absolute;
272         top: 100%;
273         left: 0;
274         width: 100%;
275         background-color: #2c3e50;
276         padding: 1rem 0;
277     }
278
279     .nav-links.active {
280         display: flex;
281     }
282
283     .nav-links li {
284         margin: 0.5rem 0;
285         text-align: center;
286     }
287
288     .hamburger {
289         display: flex;
290     }
291
292     .hero-content h1 {
293         font-size: 2rem;
294     }
295
296     .hero-content p {
297         font-size: 1rem;
298     }
299
300     .menu-grid {
301         mns: 1fr;
302
303
304     .slides img {
305         height: 250px;
306     }
307 }
```

Click to collapse the range.


Delicious Eats
Home
Menu
Gallery
Contact
View Menu


Delicious Eats
Home
Menu
Gallery
Contact
Our Menu
Pasta
Italian Pasta
Delicious handmade pasta with fresh ingredients.
$12.99
Burger
Gourmet Burger
Juicy burger with all the fixings.
$9.99
Salad
Fresh Salad
Healthy and refreshing salad options.
$7.99
Dessert
Chocolate Cake
Rich and decadent chocolate cake.
$5.99
Gallery
Interior
Food 1
Food 2
Chef



Explanation:
Copilot generated the HTML structure containing navigation links and content sections. The CSS file implemented card-based menu layouts using Flexbox. JavaScript was used to create a simple image slider that automatically transitions between images in the gallery section.

Output:
The final landing page includes:
Navigation bar with menu links
Menu section displaying food items in styled cards
Gallery with a working JavaScript image slider
Responsive layout that adapts to mobile and desktop screens.

Observation:
AI significantly accelerated UI development by generating reusable layout components. Custom animations and transitions improved the visual experience.

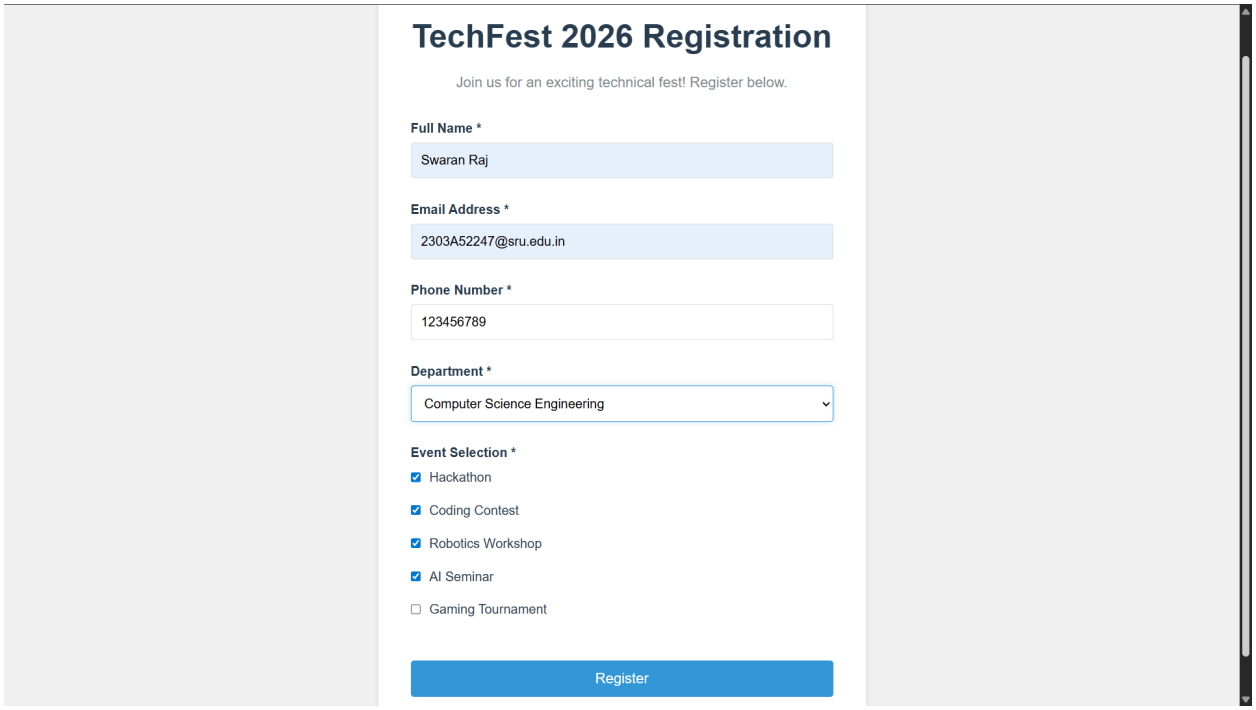
Task 3: AI-Powered Event Registration Form
Prompt:“#Generate a complete event registration form using HTML, CSS, and JavaScript for a university technical fest.

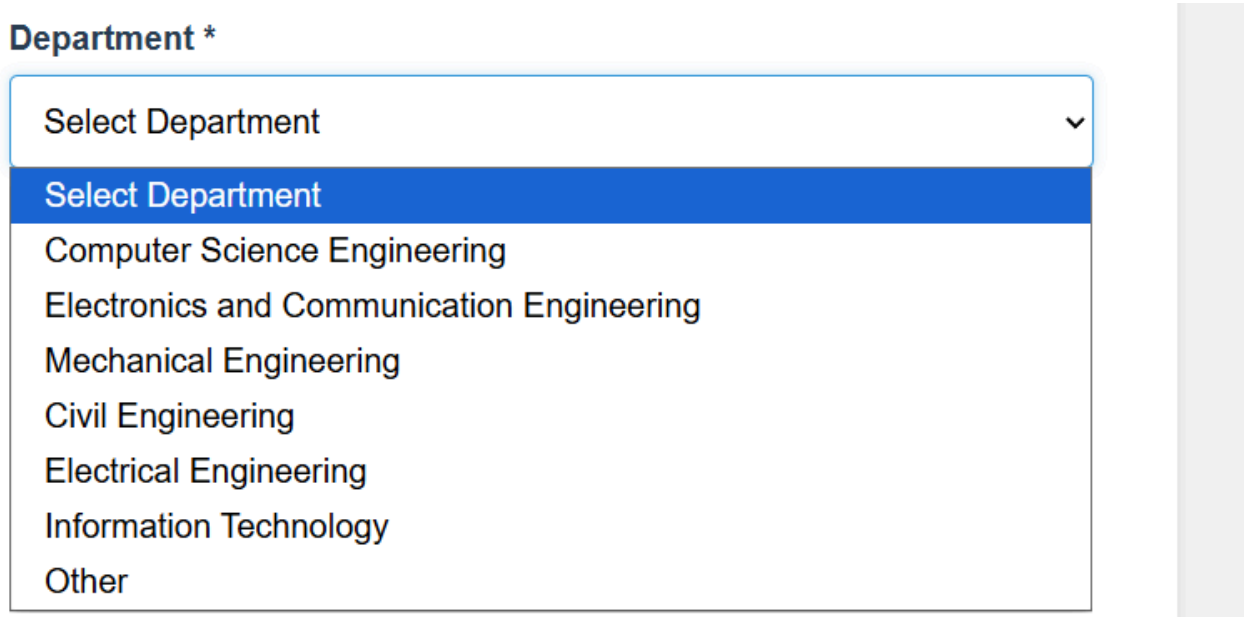
The form should include fields such as Name, Email, Phone Number, Department, and Event Selection.
Use CSS to design a clean and user-friendly interface with proper spacing and alignment.
Implement JavaScript validation to ensure the email format is correct, the phone number contains the correct number of digits, and required fields are not left empty.
Provide clear validation messages when incorrect data is entered.
Ensure the form layout is responsive and accessible on different screen sizes."

registration.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>University Technical Fest Registration</title>
    <link rel="stylesheet" href="registration.css">
</head>
<body>
    <div class="container">
        <header>
            <h1>TechFest 2026 Registration</h1>
            <p>Join us for an exciting technical fest! Register below.</p>
        </header>

        <form id="registrationForm" novalidate>
            <div class="form-group">
                <label for="name">Full Name *</label>
                <input type="text" id="name" name="name" required>
                <span class="error" id="nameError"></span>
            </div>

            <div class="form-group">
                <label for="email">Email Address *</label>
                <input type="email" id="email" name="email" required>
                <span class="error" id="emailError"></span>
            </div>

            <div class="form-group">
                <label for="phone">Phone Number *</label>
                <input type="tel" id="phone" name="phone" pattern="[0-9]{10}" required>
                <span class="error" id="phoneError"></span>
            </div>

            <div class="form-group">
                <label for="department">Department *</label>
                <select id="department" name="department" required>
                    <option value="">Select Department</option>
                    <option value="CSE">Computer Science Engineering</option>
                    <option value="ECE">Electronics and Communication Engineering</option>
                    <option value="ME">Mechanical Engineering</option>
                    <option value="CE">Civil Engineering</option>
                    <option value="EE">Electrical Engineering</option>
                    <option value="IT">Information Technology</option>
                    <option value="Other">Other</option>
```

```html
            </select>
            <span class="error" id="departmentError"></span>
        </div>

        <div class="form-group">
            <label>Event Selection *</label>
            <div class="checkbox-group">
                <label class="checkbox-label">
                    <input type="checkbox" name="events" value="hackathon"> Hackathon
                </label>
                <label class="checkbox-label">
                    <input type="checkbox" name="events" value="coding"> Coding Contest
                </label>
                <label class="checkbox-label">
                    <input type="checkbox" name="events" value="robotics"> Robotics Workshop
                </label>
                <label class="checkbox-label">
                    <input type="checkbox" name="events" value="ai"> AI Seminar
                </label>
                <label class="checkbox-label">
                    <input type="checkbox" name="events" value="gaming"> Gaming Tournament
                </label>
            </div>
            <span class="error" id="eventsError"></span>
        </div>

        <button type="submit" class="submit-btn">Register</button>
    </form>

    <div id="successMessage" class="success-message" style="display: none;">
        <p>Registration successful! Thank you for registering.</p>
    </div>
</div>

<script src="registration.js"></script>
</body>
</html>
```





Explanation:

Copilot generated an HTML form with multiple input fields and labels.
The external CSS file styled the form for readability using proper alignment and spacing.
JavaScript validation logic checks email patterns, verifies phone number length, and prevents submission if required fields are missing.

Output:

The generated system provides:
A structured event registration form
Real-time validation messages
Prevention of invalid form submissions.

Observation:
AI-generated validation scripts simplified form development while ensuring accurate user input.

Task 4: AI-Assisted E-Commerce Product Page

Prompt:"#Create an e-commerce product page using HTML, CSS, and JavaScript.
Display products in a grid-based layout using CSS Grid or Flexbox.
Each product card should include an image, product name, price, and an Add to Cart button.
Implement a JavaScript function to manage a shopping cart and display a cart item counter in the header.
Add visual feedback when an item is added to the cart to improve user interaction.
Ensure the page remains responsive and visually organized."

ecommerce.html

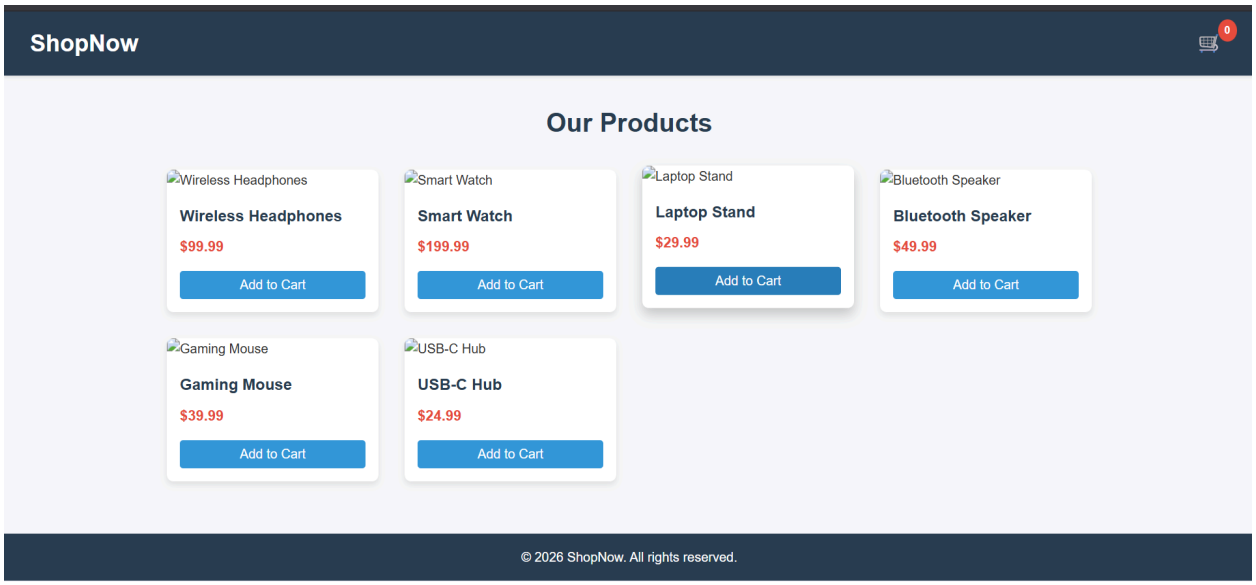
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>E-Commerce Product Page</title>
    <link rel="stylesheet" href="ecommerce.css">
</head>
<body>
    <header>
        <h1>ShopNow</h1>
        <div class="cart">
            <span class="cart-icon">🛒</span>
            <span class="cart-count" id="cartCount">0</span>
        </div>
    </header>

    <main>
        <section class="products">
            <h2>Our Products</h2>
            <div class="product-grid" id="productGrid">
                <!-- Products will be dynamically added here -->
            </div>
        </section>
    </main>

    <footer>
        <p>&copy; 2026 ShopNow. All rights reserved.</p>
    </footer>

    <script src="ecommerce.js"></script>
</body>
</html>
```

ecommerce. js

```javascript
function addToCart(productId) {
    const product = products.find(p => p.id === productId);
    if (product) {
        cart.push(product);
        updateCartCount();

        // Visual feedback
        const button = document.querySelector(`[data-product-id="${productId}"]`);
        button.textContent = "Added!";
        button.classList.add('added');

        // Reset button after 2 seconds
        setTimeout(() => {
            button.textContent = "Add to Cart";
            button.classList.remove('added');
        }, 2000);
    }
}

// Function to render products
function renderProducts() {
    const productGrid = document.getElementById('productGrid');

    products.forEach(product => {
        const productCard = document.createElement('div');
        productCard.className = 'product-card';

        productCard.innerHTML = `
            <img src="${product.image}" alt="${product.name}" class="product-image">
            <div class="product-info">
                <h3 class="product-name">${product.name}</h3>
                <p class="product-price">$${product.price.toFixed(2)}</p>
                <button class="add-to-cart-btn" data-product-id="${product.id}" onclick="addToCart(${product.id})">Add to Cart</button>
            </div>
        `;

        productGrid.appendChild(productCard);
    });
}

// Initialize the page
document.addEventListener('DOMContentLoaded', () => {
    renderProducts();
    updateCartCount();
});
```

ecommerce.css

```css
/* Reset and base styles */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Arial', sans-serif;
    line-height: 1.6;
    color: #333;
    background-color: #f8f9fa;
}

/* Header */
header {
    background-color: #2c3e50;
    color: #fff;
    padding: 1rem 2rem;
    display: flex;
    justify-content: space-between;
    align-items: center;
    box-shadow: 0 2px 4px rgba(0,0,0,0.1);
}

header h1 {
    font-size: 1.8rem;
}

.cart {
    display: flex;
    align-items: center;
    position: relative;
}

.cart-icon {
    font-size: 1.5rem;
    margin-right: 0.5rem;
}

.cart-count {
    background-color: #e74c3c;
    color: #fff;
    border-radius: 50%;
    padding: 0.2rem 0.5rem;
```



Explanation:

Copilot generated a product catalog layout using CSS Grid. JavaScript functions were implemented to handle cart operations and update the cart counter dynamically. Interactive feedback was added to indicate when products are successfully added to the cart.

Output:

The e-commerce page includes:
Product catalog grid layout
Functional Add-to-Cart feature
Dynamic cart counter displayed in the header.

Observation:

AI-assisted development enabled quick implementation of interactive features while maintaining organized front-end structure.

Task 5: Responsive Web Page Layout

Prompt:“#Generate a responsive webpage layout using HTML and CSS that includes a header, main content section, and footer.
The header should contain navigation links, while the main section should include placeholder text or images.
Use responsive CSS techniques such as Flexbox, Grid, and media queries to ensure the layout adapts to different screen sizes including

desktop, tablet, and mobile devices.

Separate HTML and CSS into different files to maintain clean project organization.

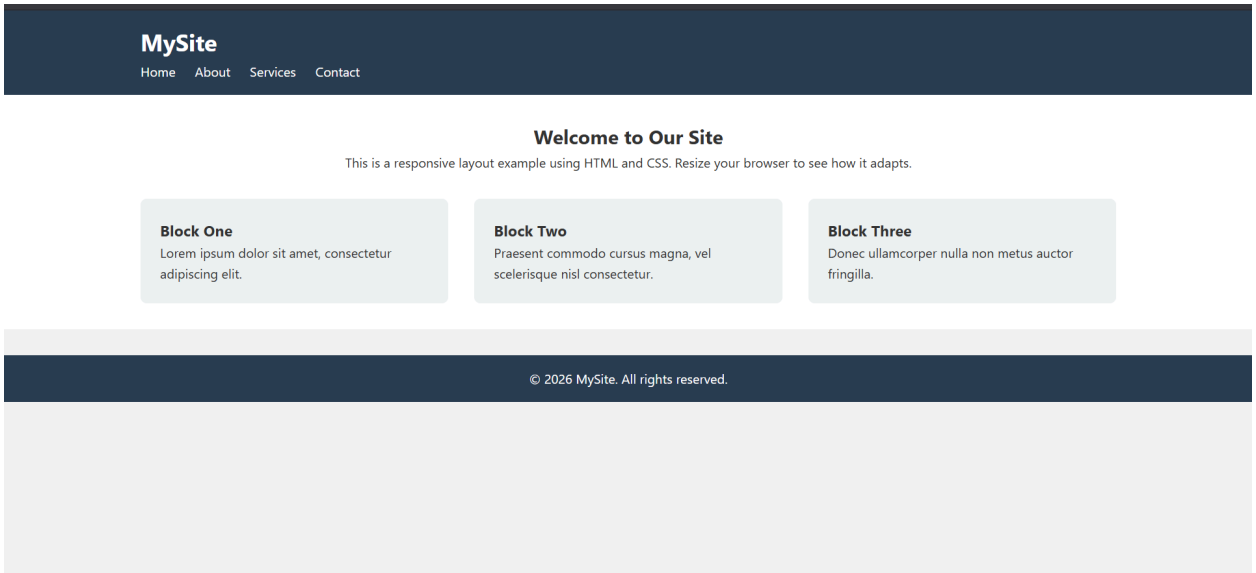
Ensure the layout is visually balanced and easy to read."

Layout.html

```html
<html lang="en">
<body>
    <header class="site-header">
        <div class="container">
            <nav class="main-nav">
                <ul>
                    <li><a href="#home">Home</a></li>
                    <li><a href="#about">About</a></li>
                    <li><a href="#services">Services</a></li>
                    <li><a href="#contact">Contact</a></li>
                </ul>
            </nav>
        </div>
    </header>

    <main class="site-main">
        <div class="container">
            <section class="hero">
                <h2>Welcome to Our Site</h2>
                <p>This is a responsive layout example using HTML and CSS. Resize your browser to see how it adapts.</p>
            </section>
            <section class="content-grid">
                <div class="content-block">
                    <h3>Block One</h3>
                    <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
                </div>
                <div class="content-block">
                    <h3>Block Two</h3>
                    <p>Praesent commodo cursus magna, vel scelerisque nisl consectetur.</p>
                </div>
                <div class="content-block">
                    <h3>Block Three</h3>
                    <p>Donec ullamcorper nulla non metus auctor fringilla.</p>
                </div>
            </section>
        </div>
    </main>

    <footer class="site-footer">
        <div class="container">
            <p>&copy; 2026 MySite. All rights reserved.</p>
        </div>
    </footer>
</body>
</html>
```

Layout.css

```css
/* Base resets */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    line-height: 1.6;
    color: #333;
    background-color: #f4f4f4;
}

.container {
    width: 90%;
    max-width: 1200px;
    margin: 0 auto;
}

/* Header */
.site-header {
    background-color: #2c3e50;
    color: #fff;
    padding: 1rem 0;
}

.site-header .logo {
    font-size: 1.8rem;
    font-weight: bold;
}

.main-nav ul {
    list-style: none;
    display: flex;
    gap: 1.5rem;
}

.main-nav a {
    color: #fff;
    text-decoration: none;
    font-size: 1rem;
    transition: color 0.3s;
```



Explanation:

Copilot generated a semantic HTML layout using header, main, and footer elements.The CSS file applied Flexbox and media queries to adjust layout alignment across various devices.This ensured that the webpage remains readable and visually organized on different screen resolutions.

Output:

The final webpage contains:
Header with navigation links
Main content section with placeholder elements
Footer containing copyright information
Responsive layout for multiple screen sizes.

Observation:

AI-assisted front-end development simplifies responsive layout design while allowing developers to refine visual elements.

Conclusion:

AI-assisted coding tools such as GitHub Copilot significantly accelerate front-end web development by generating structured HTML, responsive CSS, and interactive JavaScript functionality.These tools improve productivity while still requiring developer oversight to refine design and maintain code quality.